

SHETH L.U.J AND SIR M.V COLLEGE

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for loading the 'stats' library, reading a CSV file, and performing a paired t-test.
- Console:** Shows the output of the t-test, including the test statistic, p-value, confidence interval, and sample estimates.
- Environment:** Lists loaded objects such as 'student_data', 'StudentsPe...', and 't_test_res...'. It also shows the 'Values' of selected objects.
- Files:** A file explorer showing the project directory structure, including files like 'animal.xlsx', 'annex1.csv', and 'p1_S120.R'.

```
> # --- 1. Load Library ---
> library(stats)
>
> # --- 2. Load Data ---
> # R will automatically change spaces in column names to dots (e.g., 'Previous.Scores')
> student_data <- read.csv("C:/Users/yukta/Downloads/StudentPerformance.csv")
>
> # --- 3. Perform Paired T-Test ---
> # We compare Previous Scores vs. Performance Index for the same students
> paired_test <- t.test(student_data$Previous.Scores,
+                       student_data$Performance.Index,
+                       paired = TRUE)
>
> # --- 4. View Results ---
> print(paired_test)
```

Paired t-test

data: student_data\$Previous.Scores and student_data\$Performance.Index
t = 183.57, df = 9999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
14.06905 14.37275
sample estimates:
mean difference
14.2209

> |